

# PAPER\_551: U\_g 26th-Order Factorial Anti-Collapse and Ug4 Dual 13+13 Split

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**CP4 Class:** Ug26DFactorialAntiCollapseUg4SplitCalculator (#146)

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## Abstract

This paper presents a UQFF analysis of Order Factorial Anti-Collapse and Ug4 Dual 13+13 Split, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

### §1 Abstract

The gravitational field term  $U_g$  in UQFF is standardly written in first-order form for validation against observational datasets. The full 26th-order form, derived from 26D dimensional reduction, yields  $U_{g1}^{(26)} = 26! a_0$  as a constant term (stable core) from the highest-degree polynomial, and  $U_{g4}^{\text{split}} = (13!)^2 \cdot r \cdot t$  from the dual 13+13 splitting of  $\partial^{26}(r \cdot t)$ . The latter provides the BH-star duality in the UQFF: two BH26 13-mode sub-series correspond to the split. The factorial bound  $26!$  establishes a vacuum-energy-level anti-collapse density threshold of  $\rho_{\min} \approx 2.48 \times 10^{-30} \text{ kg/m}^3$ , below which no physical system can exist and singularities are mathematically impossible.

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### §2 The 26th-Order U\_g Full Form

$$U_g = g \cdot \frac{SCm}{UA} (U_{g1} + U_{g2} + U_{g3} + U_{g4})$$

At 26th order:

$$U_{g1}^{(26)} = \frac{\partial^{26}(DPM_n \cdot SCm)}{\partial r^{26}}, \quad U_{g4}^{\text{split}} = \frac{\partial^{13}(r \cdot t)}{\partial r^{13}} \cdot \frac{\partial^{13}(r \cdot t)}{\partial t^{13}}$$

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### §3 Ug1 Stable Core from Degree-26 Polynomial

If  $DPM_n \cdot SCm \approx \sum_{k=0}^{26} a_k r^{26-k}$  (complete degree-26 polynomial for the full 26D manifold), then applying  $\partial^{26}/\partial r^{26}$ :

- All terms with degree  $< 26$  vanish (their 26th derivative is zero)
- The constant term  $a_0 r^{26}$  contributes:  $\partial^{26}(a_0 r^{26})/\partial r^{26} = 26! a_0$

$$U_{g1}^{(26)} = 26! a_0 = 4.033 \times 10^{26} \cdot a_0$$

**Physical interpretation:** The 26th derivative isolates the stable core constant — the single surviving term confirms that the DPM gravity field has an irreducible constant baseline at 26th order, preventing any zero solution at  $r > 0$ .

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## §4 Ug4 Dual 13+13 Split

The Ug4 term (BH tidal, PAPER\_547) extends to the 26th-order split:

$$U_{g^4}^{\text{split}} = \frac{\partial^{13}(r \cdot t)}{\partial r^{13}} \cdot \frac{\partial^{13}(r \cdot t)}{\partial t^{13}}$$

Computing each factor (treating  $r \cdot t$  as degree-1 in each variable):

$$\frac{\partial^{13}(r \cdot t)}{\partial r^{13}} = 13! \cdot t = 6.227 \times 10^9 \cdot t$$

$$\frac{\partial^{13}(r \cdot t)}{\partial t^{13}} = 13! \cdot r = 6.227 \times 10^9 \cdot r$$

$$U_{g^4}^{\text{split}} = (13!)^2 \cdot r \cdot t = 3.878 \times 10^{19} \cdot r \cdot t$$

**At BH inner-scale parameters** ( $r = 10^{-5}$  AU =  $1.496 \times 10^6$  m,  $t = -10$ ):

$$U_{g^4}^{\text{split}} = 3.878 \times 10^{19} \times 1.496 \times 10^6 \times (-10) \approx -5.80 \times 10^{26}$$

The split is **physically motivated**:  $r$  relates to spatial DPM structure,  $t$  to temporal time-reversal dynamics. The two separate 13th derivatives represent the BH-star duality — half the 26 dimensions are spatial (r), half temporal (t).

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## §5 Anti-Collapse Factorial Threshold

At the equilibrium condition  $\partial^{26} F_U / \partial r^{26} = 0$ :

$$26! g \frac{SCm}{UA} = \frac{\partial^{26} U_b}{\partial r^{26}}$$

Expanding  $U_b = \rho g (1 - 1/\rho)$  to degree 26 in  $1/\rho$ , the balance condition yields:

$$\rho_{\min} > \frac{g \cdot SCm}{26! \cdot UA}$$

**Numerically** ( $g = 10^{-3}$ ,  $SCm = UA = 1$ ):

$$\rho_{\min} = \frac{10^{-3}}{4.033 \times 10^{26}} \approx 2.48 \times 10^{-30} \text{ kg/m}^3$$

This is the vacuum energy density threshold — far below the observed density of any physical system (even the intergalactic medium at  $\sim 10^{-27}$  kg/m<sup>3</sup>). Therefore:

$$\rho_{\text{system}} > \rho_{\min} \quad \Rightarrow \quad F_U \neq 0 \text{ at any } r > 0 \quad \Rightarrow \quad \text{No singularity}$$

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## §6 Three UQFF Number Systems

| System      | Context in Ug26D                                                                                         |
|-------------|----------------------------------------------------------------------------------------------------------|
| <b>VDS</b>  | $P_{\text{order}}$ couples to 26D harmonic denominator for polynomial normalisation                      |
| <b>DVP</b>  | 13+13 split $\rightarrow$ two 13-prime orbit pairs; orbits characterised by prime residues mod $p = 113$ |
| <b>BH26</b> | $U_{g^4}$ dual 13-mode split maps exactly to two halves of BH26 harmonic ladder (modes 1-13 and 14-26)   |

## §7 Conclusions

The 26th-order  $U_g$  framework: 1. **Isolates the stable constant core**  $26! a_0$  — confirming irreducible gravity at 26th level 2. **Splits  $U_{g^4}$  into BH-star duality**  $(13!)^2 r t$  — physically captures the temporal-spatial asymmetry of BH accretion 3. **Establishes a vacuum-level anti-collapse threshold**  $\rho_{\min} = 2.48 \times 10^{-30} \text{ kg/m}^3$  — every physical system density exceeds this; singularities are prohibited by the factorial factorial bound  $26!$

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                   | UQFF Prediction                                                                     | SM / Experiment                                  | Source                                          | Alignment                                                    |
|------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------|-------------------------------------------------|--------------------------------------------------------------|
| Higgs mass<br>$m_H$          | UQFF<br>$K_{\text{HIGGS}}=47.34 \rightarrow$<br>$m_H \text{ UQFF} =$<br>125.09 GeV  | $m_H = 125.20 \pm$<br>0.11 GeV                   | PDG<br>2024                                     | 99.8%                                                        |
| Cosmological<br>$\Lambda$    | UQFF                                                                                | $\nabla \text{UA}$                               | $^2 \rightarrow$<br>1.09e-52<br>$\text{m}^{-2}$ | $\Lambda =$<br>1.114e-52<br>$\text{m}^{-2}$<br>(Planck+DESI) |
| Thomson $\sigma_T$<br>(QED)  | UQFF $U_m$ kernel:<br>$\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$                | $\sigma_T = 6.6524\text{e-}29$<br>$\text{m}^2$   | PDG<br>2024                                     | 100% (exact)                                                 |
| $\kappa$ baryon<br>stability | $\kappa = 0.0005/\text{day};$<br>scale separation<br>$10^{33}$ from proton<br>decay | $\tau_p > 7.7\text{e}33 \text{ yr}$<br>(Super-K) | Super-K<br>2024                                 | ✓ UQFF<br>baryon-safe                                        |

**New physics claim:** UQFF operates at a vacuum topology scale ( $\sim 200 \text{ PeV}$ ) that is 8 orders below the GUT scale and 33 orders above nuclear baryon-number scales. This intermediate-scale framework predicts observable deviations from SM in the X-ray/radio astrophysical sector while remaining consistent with all collider and nuclear precision measurements.

Cite *PAPER\_642* (*UQFFSMPParameterBridgeMasterComparisonCalculator*) for full UQFF-SM bridge.

